

ELSIE DAVID

+1 (510) 990-4735 | elsienanbam@gmail.com | Bay Area, CA, USA | linkedin.com/in/elsienanbamdavid/ | elsienanbam.github.io/

PROFESSIONAL SUMMARY

Data scientist with experience applying Python, SQL, experimentation, causal inference, and applied machine learning (RAG and retrieval) to improve decision-making, model performance, and reporting reliability. Background spans GTM analytics, predictive modelling, and cross-functional problem-solving, with a track record of translating ambiguous commercial, policy, and operational questions into structured, decision-ready analysis.

SKILLS

Languages & Libraries:: Python (pandas, NumPy, scikit-learn, statsmodels), SQL
Methods: Predictive modeling, experimentation, causal inference, retrieval-augmented generation (RAG), LLM evaluation, feature analysis
Data & Tools: BI dashboards, exploratory analysis, data governance, Jupyter, Git/GitHub

PROFESSIONAL EXPERIENCE

W.I.N Business Solutions
Data Governance Analyst

- Analysed multi-market commercial, pricing, and operational data across 3+ regions to identify performance drivers and inform pricing, profitability, and route-to-market decisions.
- Translated dense business reviews and ambiguous client questions into structured analytical frameworks, enabling consistent metric definitions and reproducible reporting.
- Designed datasets, governance workflows, and documentation standards that improved traceability and reduced manual reconciliation by 25–30%.
- Synthesised findings into client-facing narratives and partnered cross-functionally to align on reporting needs and strengthen data usability for strategic planning.

Oakland, CA, USA
09/2025 - 01/2026

Stripe
Data Analyst Intern

- Analysed end-to-end lead funnel performance in SQL and Python, identifying high-signal lead attributes and downstream quality indicators that contributed to a 70% YoY lift in lead conversion.
- Reconstructed a legacy B2B lead-scoring model from multi-touch attribution data and ran a 90-day retrospective simulation; showed that dropping lower-signal MQL volume preserved >99.5% of marketing-generated pipeline value, reframing the change as load reduction rather than volume loss.
- Built scalable SQL pipelines, cohort views, and BI dashboards used by Go-To-Market stakeholders to monitor performance, diagnose model behaviour, and prioritise follow-up.

San Francisco, CA, USA
05/2024 - 08/2024

Heritage Bank of Nigeria · Union Bank of Nigeria
Enterprise Information Systems Administrator

- Built reporting and monitoring infrastructure for core banking systems, enabling earlier detection of operational risk and performance bottlenecks (supporting 99%+ uptime in a highly regulated environment).
- Analysed system usage, capacity, and reliability trends and partnered with engineering, operations, and compliance teams to align reporting with business and regulatory requirements.

Lagos, Nigeria
04/2020 - 07/2023

PROJECTS

Triage - Eviction Risk Prioritisation (in progress)

- Designing a calibrated supervised model on Princeton Eviction Lab and ACS Census data to help housing nonprofits prioritise outreach to high-risk census tracts.
- Scope: gradient-boosting and logistic baselines, probability calibration (Brier score, reliability diagrams), threshold optimisation, SHAP interpretability, and fairness analysis across racial and income subgroups, shipped as a Streamlit decision tool.

05/2026 - Present

Unbarred (AI Powered Local Law Guide)

- Built an end-to-end RAG pipeline to parse, chunk, embed, and retrieve information from non-searchable county ordinance PDFs, transforming unstructured legal documents into a searchable retrieval system.
- Benchmarked 6 retrieval configurations (BM25, FAISS across 4 embedding models, ColBERT) on section-match metrics and ran blind LLM comparison (Claude 3 Opus vs GPT-4). Stack: Python, FAISS, Supabase.

11/2024 - 05/2025

Public Policy Lab (Habitat for Humanity)

- Designed and executed three causal identification strategies (DiD at ZIP and home level, neighbor comparison, synthetic control with placebo testing) to evaluate the effect of housing interventions on property values in Philadelphia
- Communicated result of impact analysis and measurement limitations behind it clearly to the nonprofit partner. Stack: Python, pandas, statsmodels

01/2025 - 05/2025

EDUCATION

University of California - Berkeley
Master's, Information Management and Systems (Data Science)

08/2023 - 05/2025

University of Jos
Bachelor's, Mathematics

08/2012 - 07/2017